

Integrating Usability Testing into the Design and Development of Data Dissemination Platform for U.S. Official Statistics

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ABSTRACT

The U.S. Census Bureau serves as the leading source of statistical data about the nation's people and economy, and has the responsibility of disseminating the data for public use. To accomplish this mission, an initiative was undertaken to develop a platform that makes the data easier to find, use, and access the official statistical data effectively and efficiently. In this paper, we will describe a human-centered approach to designing and developing the platform, in particular, the process of incremental improvement in search functionality and search results presentation through usability evaluation. The process involves user research, expert review of design concept, low-fidelity wireframe usability testing, high-fidelity usability evaluation, and collaboration between the government agency and academic institutions. Our progress made so far in this project demonstrates the soundness of this human-centered approach.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **HCI design and evaluation methods**; • **Usability testing**;

KEYWORDS

Usability, Human-centered, User-centered, Data dissemination

ACM Reference Format:

Lin Wang, Zachary Whitman, Erica Olmsted-Hawala, and Robert Chestnut. 2021. Integrating Usability Testing into the Design and Development of Data Dissemination Platform for U.S. Official Statistics. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3411763.3443448>

1 INTRODUCTION

Official statistics refer to the statistical data produced by the government through survey or other methods. Official statistics are an important source of evidence for policy making [1]. For example, 132 U.S. Federal Programs in 2015 used the economic, demographic, and housing data publicly disseminated by the U.S. Census Bureau to support making decisions to distribute more than \$675 billion

dollars of federal funds [2]. Open governmental data (OGD) have long been understood to be of great intrinsic value for federal, state, and local programs and initiatives [3]. The data can also assist in civic engagement [4], drive both public and private innovation [5], and have direct economic and societal value [6].

Over the years, the U.S. Census Bureau has created several data portals, applications, websites, and application programming interfaces (APIs) as a means to provide the public with better access to the official statistics. These efforts have led to a proliferation of various data dissemination systems that were either designed as general-purpose dissemination applications (e.g., American FactFinder) or specifically tailored to a use case (e.g., Census Business Builder). However, because many of these systems were independently developed over the past three decades, there are now several different designs, approaches, and information architectures that have made it challenging for a user to find a suitable dissemination system. Furthermore, due to different user interface designs, skills learned from using one system are not readily transferrable to another. This has led to usability challenges for the data user as the data systems are often perceived as overwhelming, too complicated, or too difficult to use.

To address these issues and to improve the overall data users' experience, the U.S. Census Bureau is developing a centralized and knowledge-based data dissemination platform that is user-driven, extensible, and integrated across all the census and surveys conducted by the Census Bureau. The intention behind the platform is to simplify the user journey by consolidating the core capabilities of existing data dissemination systems into a single unified portal and API. It will not only streamline the discovery, access, and consumption of the data collected by the Census Bureau, but also facilitate users' interactions with the system.

In this paper, we describe the approach the U.S. Census Bureau has been taking in developing human-centered user interface for the centralized data dissemination system with two case studies.

2 HUMAN-CENTERED APPROACH

We integrated a human-centered approach into the platform system development lifecycle. The approach started with user research. Two types of use experience data were collected: User-reported data and behavioral observation data. User-reported data were collected through semi-structured interviews while behavioral observation data through direct observation of interviewees interacting with the Census Bureau's existing data website. Thirty one individuals, including both professional and novice users with various backgrounds, participated in the interview. The data were categorized with three aspects: (1) extent of data usage, (2) desired data search output, (3) depth of data consumption. Four types of users

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CHI '21 Extended Abstracts, May 08–13, 2021, Yokohama, Japan

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ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3443448>

emerged from the analysis: application developer, summary data user, raw data user, and ad hoc user. In addition, we engaged a private-sector contractor to conduct a similar user study, and obtained similar results. We further consolidated user types into two groups: Experienced professional users and novice users. These two groups possess different levels of data use proficiency in five aspects: (1) Seeking – locating and acquiring reliable information; (2) Visualizing – creating graphical and aesthetic representation of information; (3) Manipulating – extrapolating data, creating new sets of data, or making intermediate calculations prior to analysis; (4) Programming – automating data use through APIs or other forms of development; (5) Analyzing – summarizing data patterns or identifying trends.

Based on the findings from the user research, the design team proposed the following design principles: (1) provide guidance for people on which tools to use, through a holistic information architecture redesign of the homepage; (2) implement intelligent corrections in semantic search and suggestions to handle human errors in navigating through the user interface; (3) provide more visually consumable data; (4) use conventional information architecture research methods on results-related content items; (5) leveraging marketing tactics on www.census.gov homepage and other channels to convey the extent of the U.S. Census Bureau's data.

The development plan made a special emphasis on usability evaluation in an attempt to maximize positive user experiences. Iterative usability evaluations were planned throughout the lifecycle of the product design and development. Specifically, the plan established the following objectives: (1) minimize potential usability problems during application design phase; (2) identify and address usability problems during application development phase; (3) assess application's usability upon the completion of development; (4) establish user performance pattern over time.

In the following, we will present two case studies of improving data user experiences through usability evaluation: Data search and search results presentation.

3 IMPROVING DATA SEARCH

In the process of developing the data search module, we implemented iterative usability testing to facilitate incremental design improvement.

3.1 Initial Search Design Assessment

The initial screen design of the search module included three areas: Left, middle, and right (Figure 1). In the middle of the screen, a list of search fields, derived from the Census Bureau's internal data structure, was presented vertically from the top all the way down through the screen. There were various search field formats: some with write-in areas, some others in dropdown format, etc. The area on the left side of the screen was used to show the "filters" that had been applied during a search, while the right side of the screen displayed instructions on how to perform a search.

We conducted a usability testing on this original design. Twenty participants (10 male and 10 female, age range: 20 - 71) took part in the testing. The participants were categorized as "novice" data users. A novice user was defined as a person who had little knowledge of or experience in using data provided by U.S. Census Bureau.

The participants performed the same 5 search tasks throughout the entire iterative cycle of usability testing (from the initial test through Round 4), so that we could compare the usability improvement or degradation through the lifecycle. The tasks were designed to assess participants' ability of using the dissemination tool to obtain information on a single topic or a single topic by a single geography; to compare a topic between two geographies; to filter information using a map, an industry topic, and/or information by year.

The following usability issues were observed during the test: (1) Ambiguous search parameters requirement. Users thought they had to fill every search parameter field before running a search, which was actually not correct. It was difficult for novice users, who had little knowledge about the inner structure of the Census Bureau's data, to come up with a set of search parameters that satisfy the logic built inside the search module. Consequently, such search often failed. This observation suggests that public users, who are not familiar with the Census Bureau's data, have a different data model in their mind. (2) Search in natural language. Users had a tendency to conduct a search in a way like a "Google search," i.e., using natural-language-like phrases and sentences. Some users actually said "I would go to Google." (3) Inappropriate information placement. If a user searched on a single keyword, as soon as the word was recognized by the search module (e.g., poverty), the word would disappear from the topic search field but reappear in the selected filters section to the left side of the screen. However, the user often did not notice its reappearing in the left side of the screen and thought the tool was broken because search terms kept disappearing out of the search field. Consequently, users could not understand how this search works.

3.2 Round-2 Search Design Assessment

The second round of usability testing was conducted on the new user interface design shown in Figure 2. The purpose of this testing was to (1) assess if the new design address the previous usability problems, and (2) to identify additional issues. Twenty novice-data-user participants (9 male and 11 female, age range: 20 - 71) participated in the testing.

User performance was clearly improved. Participants knew immediately how to begin searching with natural-language search queries. However, a functional problem was discovered: The back end of the search module was not sufficiently comprehensive to parse out some of the natural language that the general user population would use when conducting searches. For example, the dissemination tool would not return relevant results to the search query "number of grocery stores in USA." After searching on that query, the top search result showed "unweighted sample count of the population." Participants would have to back up and try another natural language search query, which might likely fail for the same reason.

Acting upon the feedback from the usability testing, the development team revised the back-end algorithms and increased natural language tagging of the Census Bureau's data so that the search terms could be better mapped onto the data elements. In addition, the development team also updated the graphic design of the front page with a more modern touch as shown in Figure 3

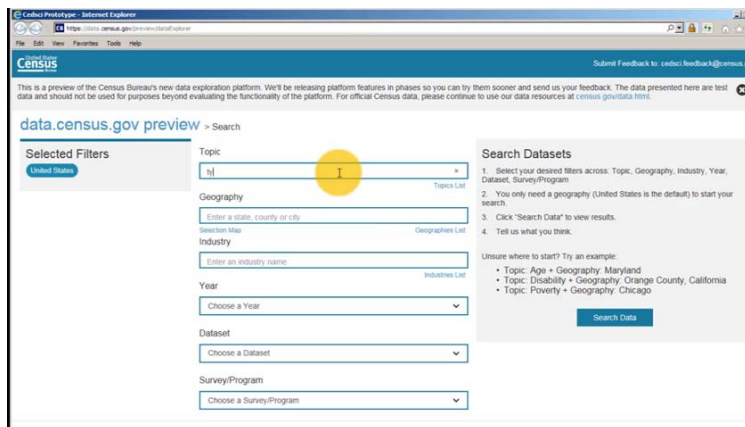


Figure 1: The origiBased on the findings from the usability testing, the design team revised the user interface with a single search field, as shown in Figure 2

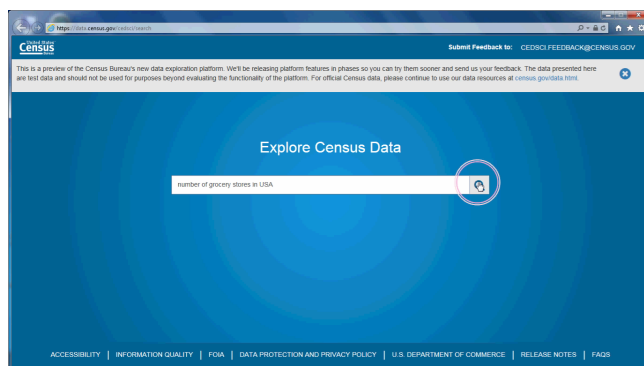


Figure 2: The first revision of the search module user interface. nal user interface design of the data search module.

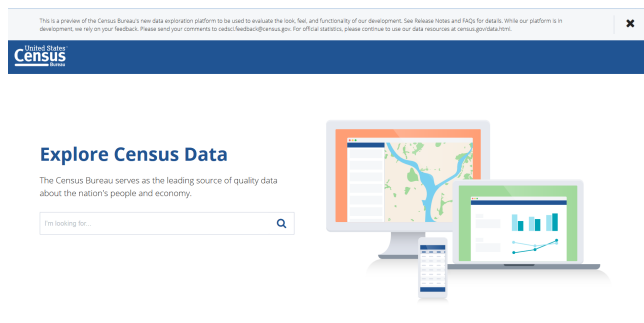


Figure 3: The Second revision of the search module user interface, with enhanced search algorithm and modern graphic design.

3.3 Round-3 Search Design Assessment

The third round of usability testing was conducted on two types of device, laptop computers and smartphones. The purpose of this test round was to assess if the dissemination tool meet the design requirements of working on large- and small-screen devices. The

user group for both devices were novice users and there were 10 participants recruited for each device type.

The main finding from this round of testing concerned the pre-populated text in the search field. It was observed that on a mobile device, the pre-populated text was noticeable but confusing. Participants thought they had to click search just at the right time after a word had appeared. Others thought they had to wait for their search term to appear.

Based on the usability observation and the recommendation from the usability team, the design team modified the design by eliminating the pre-populated text. This modification proved successful as users could understand how to interact with the search after the modification.

3.4 Round-4 Search Design Assessment

In the fourth round of usability testing, we compared the search performance between two different user groups: novice users and expert users. An expert user was defined as one who had knowledge of and experience in using data provided by U.S. Census Bureau.

The testing results showed that both groups of participants could easily start with a single-field search which was the default search setting. On the other hand, the expert users found the single-field search inconvenient for performing a more complex search that involved using logical operators (e.g., union) and/or including or excluding certain terms. In addition, we observed that when on mobile devices, users sometimes had a tendency to use mobile shortcuts, such as searching with terms “near me” to harness the GPS capabilities ubiquitous to mobile searches.

The search module design was revised once more to add the advanced search function and the function of GPS location. The new design placed the links “Search my location” and “Advanced Search” just beneath the single search field on the front page (Figure 4). To further improve user experience and search efficiency, the new design also included predictive text that would appear as the user started to type in the search field. The predictive text also included keywords that the U.S. Census Bureau would like to promote as

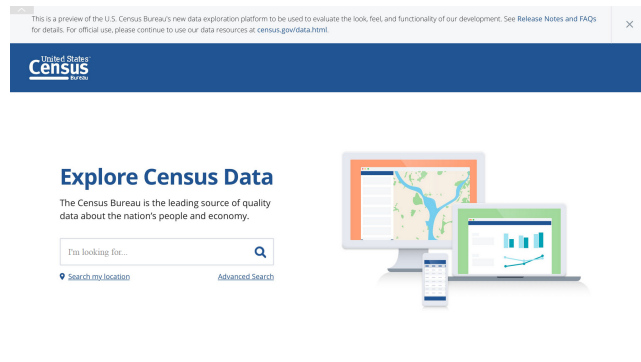


Figure 4: The Fourth revision of the search module user interface, with the addition of two links, “Search my location” and “Advanced Search.”

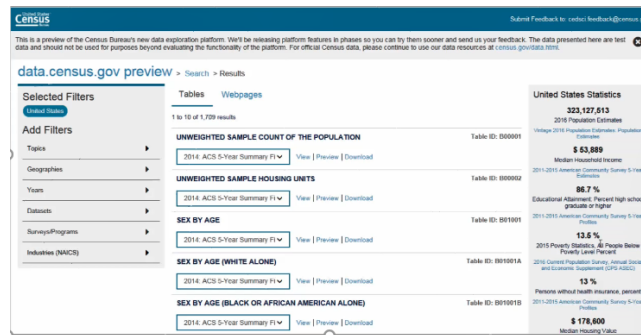


Figure 5: A sample screenshot of the initial user interface design for the search results presentation.

a way to facilitate relevant content that the general users may be seeking.

4 IMPROVING SEARCH RESULTS PRESENTATION

The search results presentation module (SRP) is another critical component of the data dissemination platform. The SRP is the interface for the data user to acquire the information they are seeking. The information they receive from SRP will directly affect the quality of their usage of the Census Bureau’s data.

4.1 Initial SRP Design Assessment

The initial design divided the SRP screen into three vertical panels: Left, middle, and right. The middle panel displayed search results, left panel search filters, and right panel search-related-data at national level. Figure 5 presents a sample screenshot. The purpose of the usability testing was to identify major usability issues. The initial usability testing was conducted on the same participants as in the initial search design assessment.

The testing results showed that users struggled to find relevant information on the SRP screen. The issues are summarized in three categories: (1) Inconsistent data presentation, e.g., the results returned from search often did not map with the query the users had just entered into the search; (2) lack of clarity in data layout

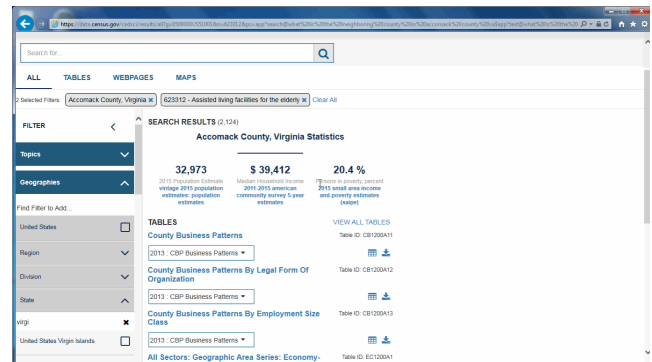


Figure 6: First revision of the user interface design for the search results presentation.

on the screen, e.g., the information on the left side of the screen was confusing and often ignored by the user, the titles of different data tables sometimes appeared so similar that it was difficult to scan and identify the right information; (3) unfamiliar terminology, e.g., many of the words displayed on the SRP screen were technical jargon which was difficult to be comprehended by data novice user.

Based on the findings, the usability team recommended that (1) SRP algorithm be enhanced with similes that could map keywords that users frequently used with the internal proprietary terminology; (2) the table titles be more informative in plain language; (3) the filter information in the left panel be more visually salient to attract users’ attention.

The SRP module was revised upon the recommendations by the usability team. The new design featured a modified filter panel on the left of the screen. Filters that had been added were also displayed across the top of the screen. Important summary data were shown at the top of the SRP screen while the rest of the results were listed below. Though table titles were not yet changed, new iconography was added to facilitate viewing and downloading the data table. The new design also incorporated a function of refining search or starting a new search from the SRP screen. Figure 6 shows the look and feel of the new design.

4.2 Round-2 SRP Design Assessment

The second round of usability testing of the revised design was conducted on the same participants as in the Round-2 search design assessment.

The testing revealed three new usability problems. (1) Inconsistency between the search query entered on search screen and the search query displayed on the SRP screen. At the time of this new design release, the single-field search with natural-language search query had already been implemented. However, the natural-language search query was replaced with “jargon” terms in the filters panel on the SRP screen. The participants would often look at the results and wonder why none of the terms they had searched were showing up on the results page. (2) Lack of information clarity. Specifically, the filtering information along the left-hand side of the screen was still confusing to many novice users; placement of the filter keywords on the top of the screen did not help the user better understand what was being applied. (3) Placement of summary data.

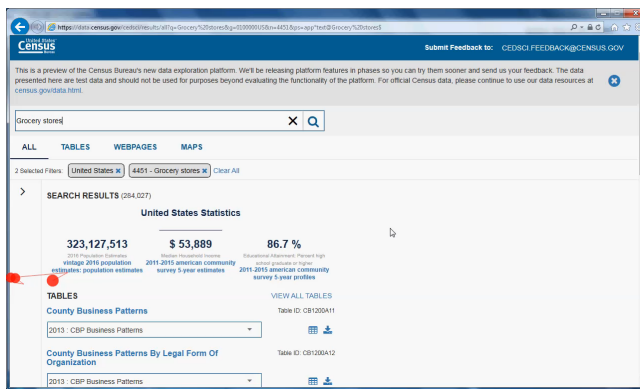


Figure 7: The second revision of the search results presentation module, with enhanced mapping from search term filter display on SRP screen. This screen displays the results from search on grocery stores. The top part of the results shows county business patterns that are correct results but not comprehended by the user.

In response to the feedback that participants didn't really notice the highlighted numbers along the far right column, the new design moved summary data to the top of the screen. However, this design often confused users who didn't read the data attentively and would think that the number was the answer to their search query, even if it wasn't. Other users saw the key numbers at the top and would think that the tool had not worked because the same number kept appearing at the top of the screen, regardless of what the user had searched on.

Based on the findings from this usability testing, the usability team recommended that the terms that were typed into the search box appear on the SRP screen as well. The usability team also concluded that filters along the left side and top didn't appear to be helpful to the users and needed to be revised. In response, the design team came up with a default view that hid the filters and made revision on the back end to address the issue of keyword searches not mapping to any Census Bureau's data. The new design of the user interface is shown in Figure 7

4.3 Round-3 SRP Design Assessment

The third round of usability testing was conducted on the same participants as Round 3 search design assessment.

Testing results showed incremental improvements while many of the same issues remained. There were two major issues: (1) Less-comprehensible results presentation - as shown in Figure 7, the participant did a search on "Grocery Stores," which returned results on SRP screen with the top portion of the result being "County Business Patterns." The link to County Business Patterns would actually lead the user to the correct table for data on Grocery stores. However, users did not know that by reading the label, as the label did not mention grocery stores. As in previous rounds of testing, the table title links failed to show pertinent information to the intended search. (2) Same usability issues existed on mobile devices and even worse due to the smaller screen. In addition, icon for downloading on the mobile phone did not follow the convention of

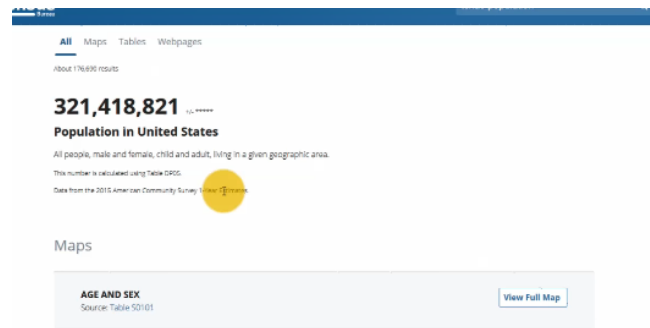


Figure 8: The third revision of the search results presentation module, with filters completely removed, the result preview taking up the entire page and smaller tabs at the top to focus on maps, tables or webpage results.

android or iPhone, which often confused the user when perform a downloading task.

The usability team recommended that (1) the development team enhance the back end so that results presentation be consistent with the corresponding search query, and (2) SRP be optimized for the smaller screen and the form factors follow Android and iPhone designs. In response to the recommendations, the design team removed the filters and opted to highlight only one result element (one number) at the top of the screen. The results section was broken into sections that users could tab through at the top of the screen (maps, tables, webpages), and these results presentation was expanded to the entire SRP screen, with preview of other screens, e.g., Maps as shown Figure 8

4.4 Round-4 SRP Design Assessment

In the fourth round of testing, we assessed whether expert users could understand better than novice users about how the layout of the results page worked. The same participants as in the Round 4 search assessment participated in the testing.

We observed that, while expert users were more comfortable with the SRP, these users complained about the same usability problems as the novice users. For example, the number at the top of the screen being not "correct" enough to be meaningful, the search query and what being returned in the SRP screen did not appear to match. On the other hand, the data experts were more savvy with the table titles, and knew a bit of the terminology that the Census Bureau uses on its products. Nevertheless, the experts stated that the table titles and terminology could be improved to include more explicit language about what was actually on the table or map.

5 CONCLUSION

This paper presents two case studies of incorporating iterative usability testing into an agile development life cycle, using the methodology of human-centered design in a government operational setting. This study demonstrates the importance of involving usability evaluation in the earlier stage of design phase and throughout the development lifecycle. It has a significant implication on quality and efficiency improvement, and on cost saving.

This continuous and multi-year process of having real world users of the Census Bureau's products interact and work with the latest version of the website tool gave the design, development, and usability teams a better understanding of what was working and what needed to be improved for a more optimal user experience. In successive iterations, we were able to identify issues, recommend solutions, and test the product again in the next release. We obtained continuous feedback with users throughout the development of the product. We will continue to refine our methodologies, particularly in user experience measurement, to provide more informative and generalizable usability assessment of data dissemination tools.

ACKNOWLEDGMENTS

The authors thank all members of the design, development, and usability testing teams for their contributions to the platform design, development and user experience assessment.

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